Unit 4, Lesson 10: Writing Equations of Parallel Lines

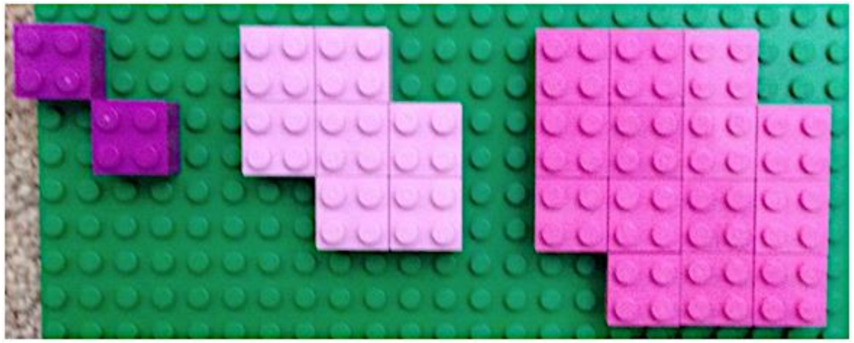
Benchmark

MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.

Learning Targets

- ☐ I can write the equation of a line that is parallel to a given line through a given point.
- ☐ I can write the equation of a line that is parallel to a given line when given two points.
- ☐ I can write the equation of a line that is parallel to a given line when given a point and the slope.

4.10.1 Warm-Up: Lego Pattern



1. What do you notice about the pattern?

2. Draw the next set of blocks in the pattern.

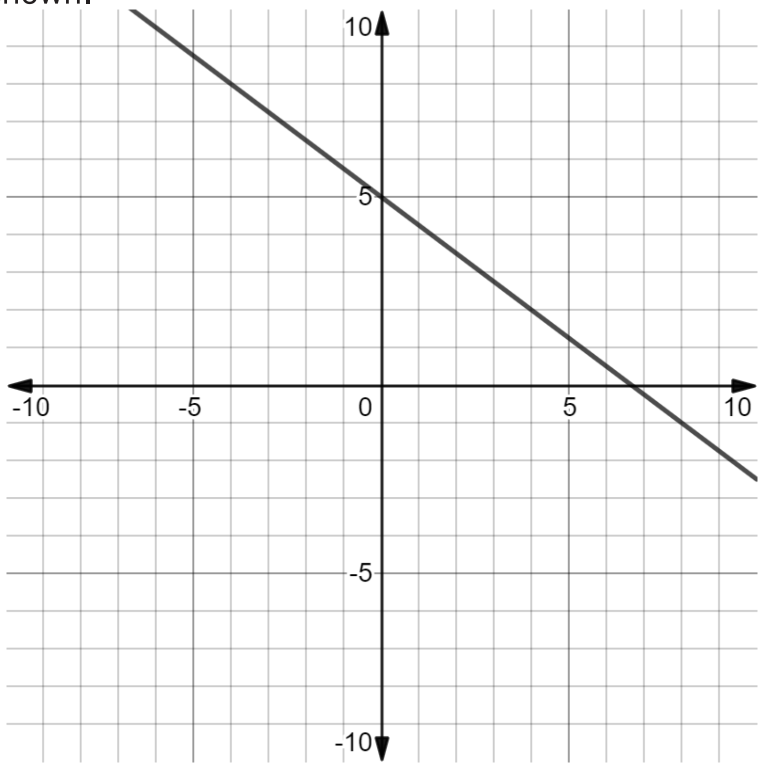
3. Complete the two different ways of writing the pattern.

Number of Blocks	$2, 7, 14,$ ____
Number of Circles	$8, 28, 56,$ ____

4.10.2 Collaboration: Parallel Lines

Lines are **parallel** if the slopes of the lines are the same.

1. The graph of the linear relationship $y - 11 = -\frac{3}{4}(x + 8)$ is shown.

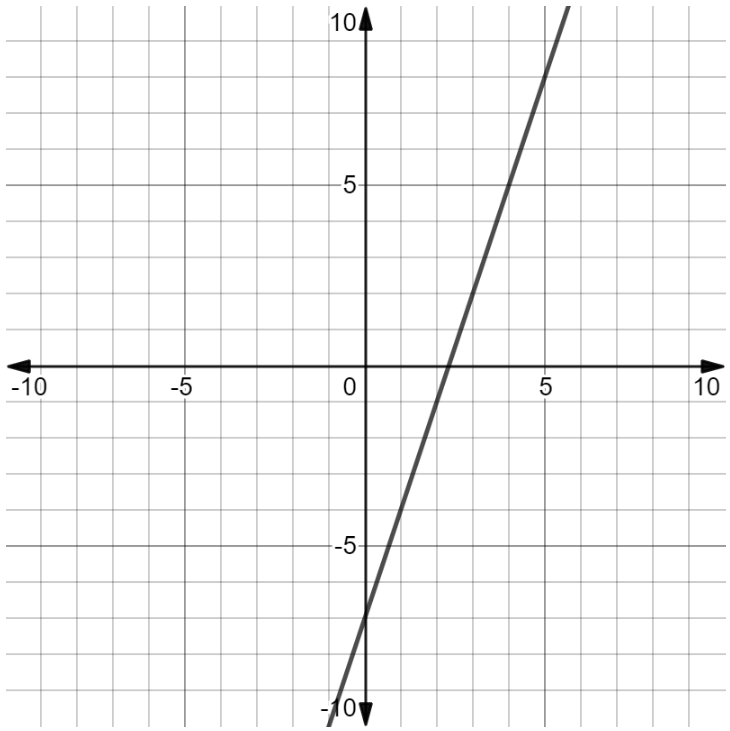


   a. Plot the point $(-1, 3)$. Then draw the line that passes through the point that is parallel to $y - 11 = -\frac{3}{4}(x + 8)$.

   b. What is the slope of the line that passes through the point $(-1, 3)$?

   c. Work with your partner to write the equation of the line that is parallel to $y - 11 = -\frac{3}{4}(x + 8)$ and passes through the point $(-1, 3)$.

2. A linear relationship is shown on the coordinate grid.



   a. Complete the equation of the line that is parallel to the line and passes through the point $(4, -6)$.

$$y + 6 = \_\_\_\_(x - 4)$$

b. Discuss with your partner why the point-slope form of a line is the easiest form of a line to use when writing the equation of a parallel line.

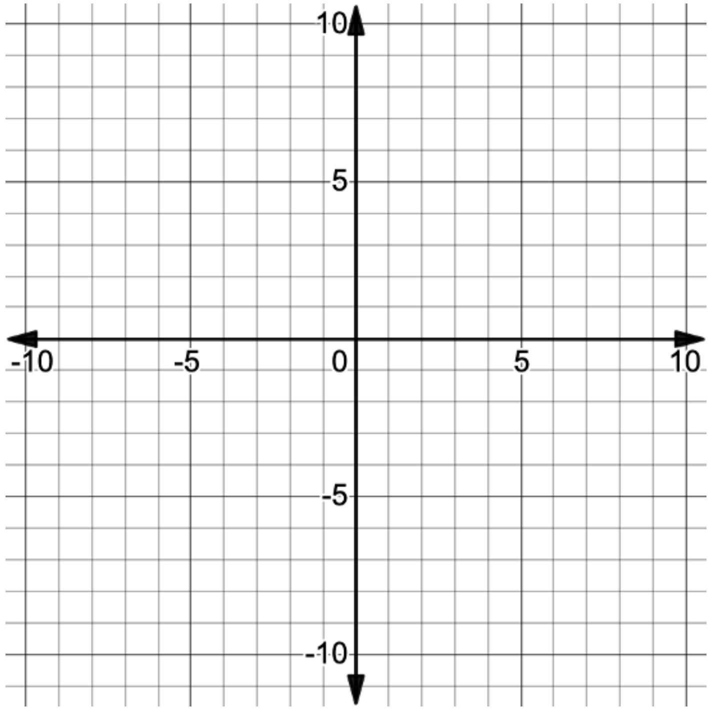
c. Write the equation in slope-intercept form.

$y =$ ________ $x +$ ________

d. Write the equation in standard form.

________ $x +$ ________ $y =$ ________

3. Plot the two points $(-2, 7)$ and $(1, 4)$ on the coordinate grid.



a. Write the equation of the line that is parallel to the line that passes through the points $(-2, 7)$ and $(1, 4)$ and goes through the point $(0, -3)$.

b. Complete the following about the equation of the line.

The reason I wrote the equation in

- ○ point-slope
- ○ slope-intercept
- ○ standard

form is because

____.

c. What method did you use to find the slope of the line?

○ I used the two points $(-2, 7)$ and $(1, 4)$ and subtraction to calculate $\frac{change\ in\ y}{change\ in\ x}$.

○ I used the graph and counted to find the $\frac{change\ in\ y}{change\ in\ x}$.

4.10.3 Guided Instruction: Parallel Lines

1. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(-5, -2)$ and $(-3, 2)$ and goes through the point $(7, -1)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

2. Write the equation of the line that passes through $(4, -7)$ and is parallel to $y = -1$.

3. Discuss with your partner how to remember that the slope of a horizontal line is 0.

4. Write the equation of the line that passes through $(4, -7)$ and is parallel to the line that passes through the points $(-1, -2)$ and $(-1, 6)$.

5. With your partner, discuss how to remember that the slope of a vertical line is undefined.

6. Complete the table.

Type	Sketch	Slope	Equations of Parallel Lines
Horizontal Line			
Vertical Line			

7. Write the equation of the line, in all three forms, that is parallel to the line $-12x + 15y = 45$ and goes through the point $(-9, 14)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

4.10.4 Your Turn!

1. Write the equation of the line that passes through $(-3, 8)$ and is parallel to $x = 4$.

2. Write the equation of the line, in all three forms, that is parallel to the line $14x - 2y = 51$ and goes through the point $(3, -11)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

4.10.5 Wrap-Up: Cool Down

1. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(-6, 8)$ and $(-2, -4)$ and goes through the point $(3, -5)$.

Point-Slope Form	Slope-Intercept Form	Standard Form